

Michigan State Plane Coordinate System Update MSPS 2022 Datum Committee

When the National Geodetic Survey (NGS) undertook the creation of the North American Datum of 1983 (NAD83), they used the best science known at the time. Geodetic experts from around the world came to the U.S. to share their expertise on that massive undertaking. Early on, NGS had decided that the new datum would be a geocentric datum – coincident with the center of mass of the earth. This was important because of the realization that the future of surveying measurements would rely heavily on space positioning techniques. They also assumed the plate velocities were fixed. One needs to realize that Global Positioning System (GPS) measurements were not utilized in the NAD83 adjustment. Subsequently, we now know that the NAD83 definition was over 2 meters off from the best estimate of the center of mass and that the plates are moving and causing non-zero changes in the crust. Over the intervening years, NGS has had to rely on a few realizations or adjustments to the NAD83 datum. Also, NGS relies on the International Terrestrial Reference Framework (ITRF) for a global view of the earth.

The NGS periodically develops long range strategic plans for identifying activities for the future. In 2013 they prepared a document outlining goals for the next decade. It was updated in 2019 with the publication of the National Geodetic Survey Strategic Plan 2019-2023 – Positioning America for the Future (https://www.ngs.noaa.gov/web/about_ngs/info/mission-strategic-planning.shtml).

NGS has since defined a new 2022 datum called, for the conterminous US, the North American Terrestrial Reference Frame of 2022 (NATRF2022). Due to this new datum, state plane coordinates throughout the country will also change. These will be referred to as the State Plane Coordinate System of 2022 (SPCS2022).

After the NGS announcement, MSPS created the 2022 Datum Committee to determine how this affects surveying and mapping activities within the state. The committee reviewed the policies and procedures established by NGS to see what the Society needed to do to prepare for the change. NGS had developed rules that the states had to follow in defining the new state plane coordinates. An important rule was that any changes to the SPCS that a state wanted would have to be supported by all the stakeholders within the state. Because of this, MSPS and the Michigan Department of Transportation (MDOT) created a Stakeholders Committee to provide additional input into the new SPCS.

In designing the new state plane coordinates NGS has said that each state will have one single state-wide zone. In Michigan, we have the Michigan GeoRef System (see Figure 1) that uses an Oblique Mercator projection reportedly modeled after Ralph Moore Berry's Hotine Skew Orthomorphic Projection. NGS has designed an Oblique Mercator projection closely aligned with the Michigan GeoRef system. One of the benefits here will be that NGS will recognize and maintain this new oblique projection whereas they have not done so up until now. This single state-wide projection will meet the needs of agencies like the Department of Natural Resources and others who have assets distributed throughout the state.

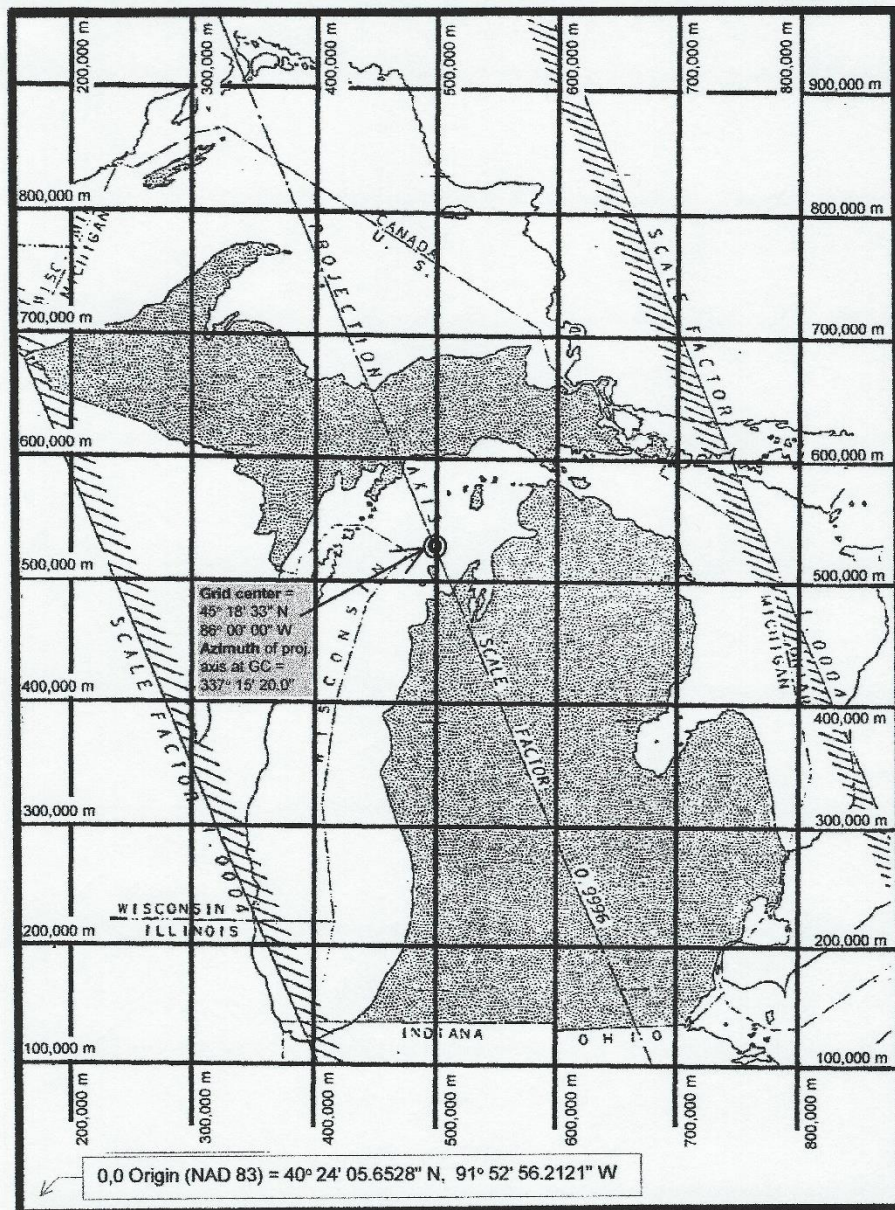


Figure 1. Michigan GeoRef from "A Primer on Projection Systems Commonly Used in Michigan", David Lusch, Fundamentals of GIS Workshop, April 2000.

The NGS design for the Oblique Mercator projection covering the state is shown in Figure 2. One can see that the distortions are significantly smaller than those that users of Michigan GeoRef have encountered.

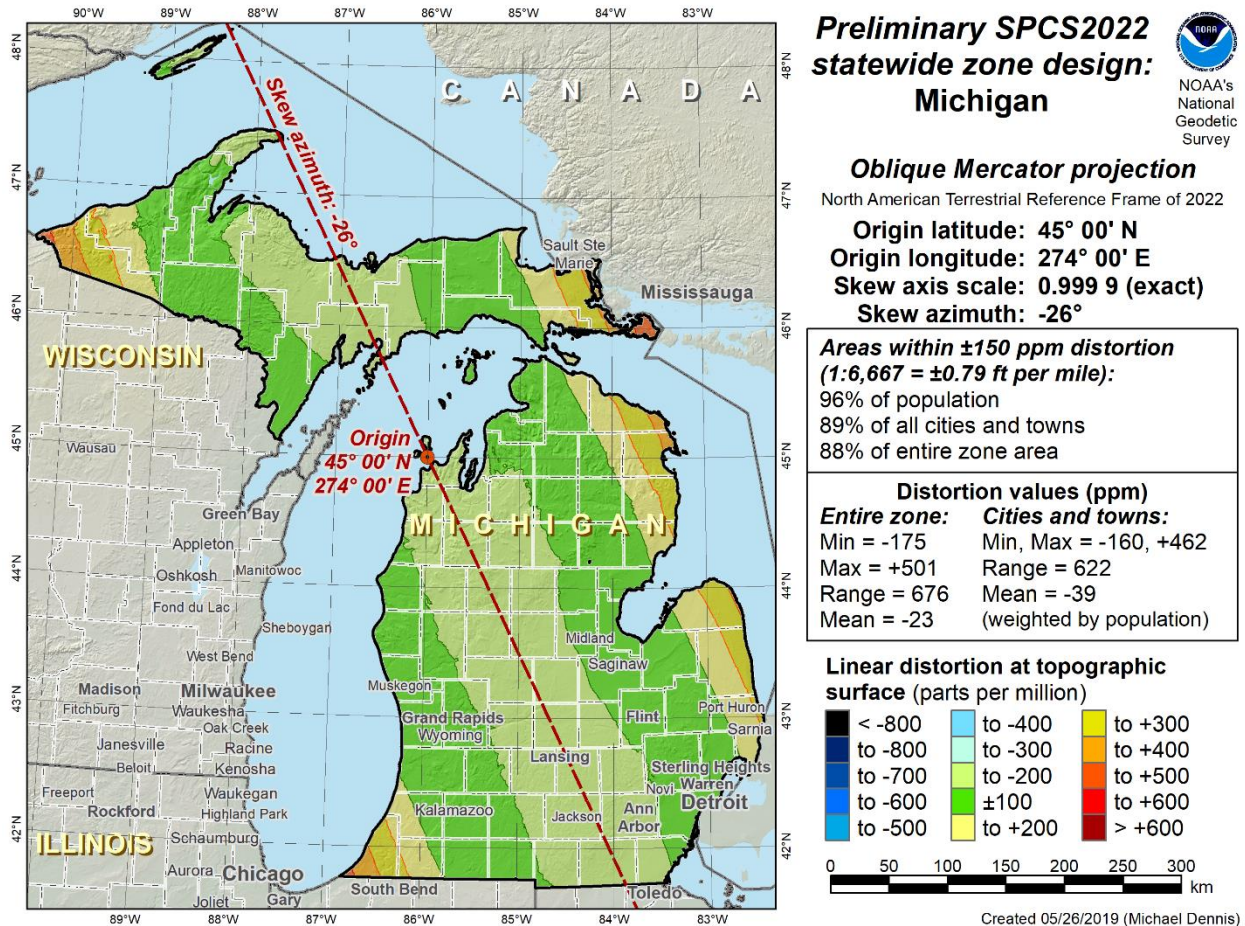


Figure 2. Michigan Oblique Mercator Projection to be developed by NGS. Note that since being created by Dr. M. Dennis, a couple of minor modifications have been applied to this graphic.

NGS had said that the existing SPCS from 1983 would be updated by them using the same configuration. This meant that Michigan could retain the same 3-zone configuration that we currently have. The committee looked at other states like Wisconsin and Indiana and determined that it would be beneficial to the surveyors in the state if we were to adopt a Low Distortion Projection (LDP) configuration for Michigan. The 2022 Datum Committee looked at a couple of possible scenarios and came up with the proposed design shown in Figure 3. The committee retained the Lambert Conic Conformal projection and applied this to all zones. The advantage of this Lambert projection is that distortions would accumulate north-south allowing for the east-west size to be larger. We finalized the design of arranging counties as a series of tiers, each being approximately 2 counties north-south and 4 counties east-west. The width was restricted in order to hold the convergence angle to a smaller value at the zone extremities. In the Upper Peninsula, the configuration is slightly different because of the size and shape of the counties.

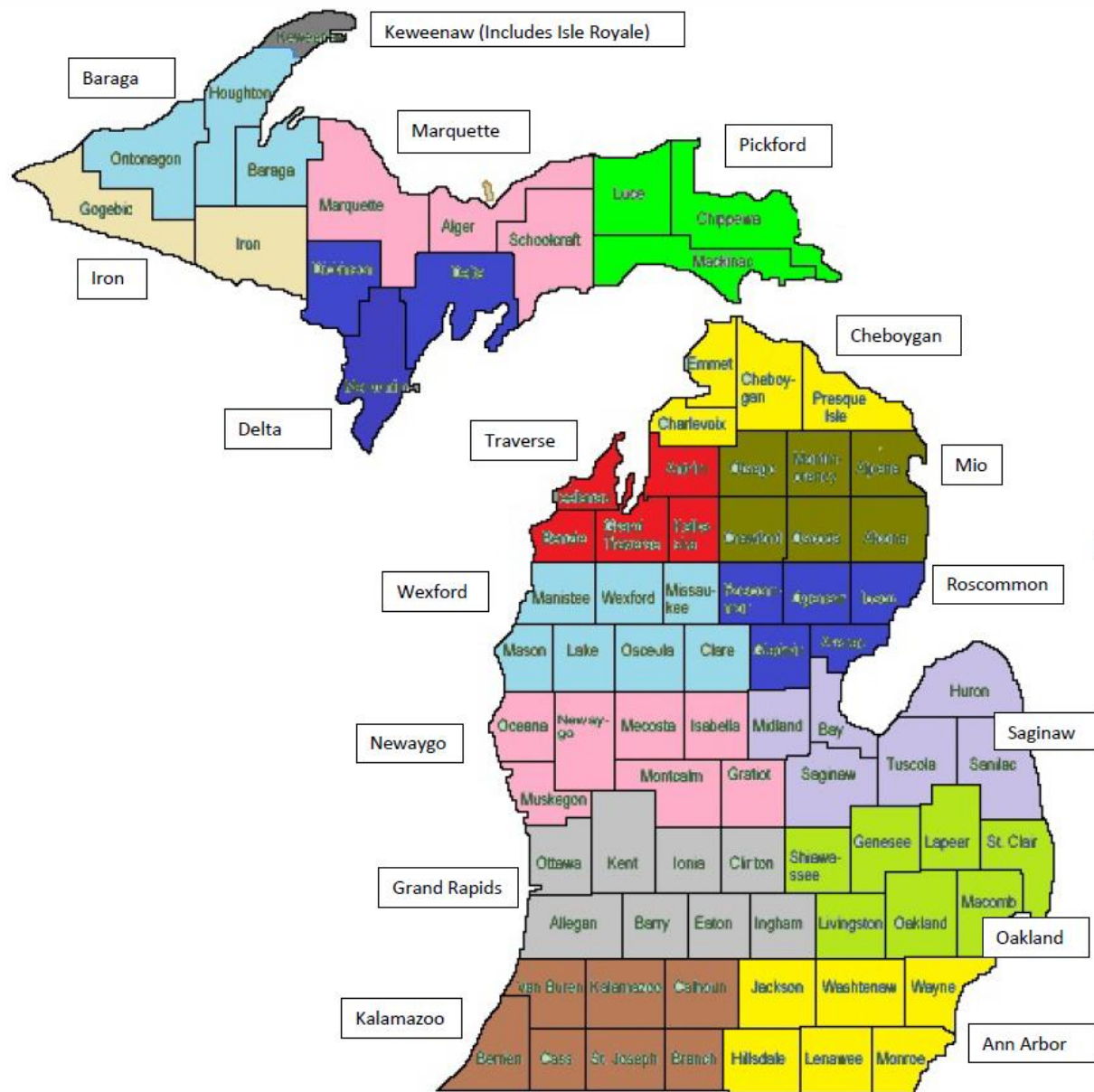


Figure 3. Proposed Low Distortion Projection zones for the State of Michigan. The zone names are identified in the boxes.

The zones will encompass the following counties:

Zone Name	Included Counties
Ann Arbor	Jackson, Washtenaw, Wayne, Hillsdale, Lenawee, Monroe
Baraga	Ontonagon, Houghton, Baraga
Cheboygan	Emmet, Charlevoix, Cheboygan, Presque Isle
Delta	Dickinson, Menominee, Delta
Grand Rapids	Ottawa, Kent, Ionia, Clinton, Allegan, Barry, Eaton, Ingham
Iron	Gogebic, Iron
Kalamazoo	Van Buren, Kalamazoo, Calhoun, Berrien, Cass, St. Joseph, Branch

Keweenaw	Keweenaw including Isle Royale
Marquette	Marquette, Alger, Schoolcraft
Mio	Otsego, Montmorency, Alpena, Crawford, Oscoda, Alcona
Newaygo	Oceana, Newaygo, Mecosta, Isabella, Muskegon, Montcalm, Gratiot
Oakland	Shiawassee, Genesee, Lapeer, St. Clair, Livingston, Oakland, Macomb
Pickford	Luce, Chippewa, Mackinac
Roscommon	Roscommon, Ogemaw, Iosco, Gladwin, Arenac
Saginaw	Midland, Bay, Huron, Saginaw, Tuscola, Sanilac
Traverse	Leelanau, Benzie, Grand Traverse, Kalkaska, Antrim
Wexford	Manistee, Wexford, Missaukee, Mason, Lake, Osceola, Clare

The purpose for embracing a low distortion projection is to create a system where, for most normal surveying applications, the combined elevation and scale factors can be ignored. Thus, one should expect to see that inverse distances derived from GPS measurements on 2 points should be essentially the same as the distance measured between the 2 points with an EDM/total station. This will alleviate problems where the combined factor is either presently ignored or applied incorrectly. As a reminder, we acknowledge that there are no projections that are completely distortion free and, therefore, high precision surveys may still have to account for these distortions. This concept of the LDP has been embraced by the Stakeholder Committee. The proposal was submitted to NGS and has been approved.

Since we have proposed an LDP, it is now the 2022 Datum Committee's task to come up with the zone parameters. Currently that effort is being headed up by Jim Leitch. We also need to determine what legislation needs to be changed because of this new datum. Finally, the committee will look at what the effects that the vertical component will have on surveying and mapping in Michigan.

The committee members include: Karl Brandys, Bob Burtch, Derek Cusimano, Brett Dodge, Larry Feindt, Max George, Jim Leitch, Chuck McCready, John Rauser, Brian Reynolds, Jeffrey Schepers, and John Ellingson (NGS – now retired). If anyone is interested in joining this effort, please contact the Central Office.